

L1.1b Transformations

Recall: L1.1a Functions + Catalog

Name _____

Solo – 9 minutes

1. Name the family, then give the domain in **both** interval and set-builder notation.

(a) $f(x) = \sqrt{x-3}$ (b) $g(x) = \frac{1}{x^2-2x}$ — also give its VA@ and HA.

2. $f(x) = x^2 - 3x$. Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$.

3. $y = \sqrt{x}$ has $D = [0, \infty)$. Write the domain of $y = \sqrt{x} + 2$ and of $y = \sqrt{x+2}$.

One of them moved the domain. Which way did it move — and is that the direction the +2 led you to expect?